

Consider:

$$X'' = -\lambda X, \quad 0 < x < 1, \quad X'(0) = aX(0), \quad X'(1) = -bX(1),$$

where $a, b \geq 0$. Without solving for the eigenvalues, show that all eigenvalues must be greater than or equal to zero. (Hint: Let $L := -\frac{d^2}{dx^2}$; assuming $Lu = \lambda u$ for a nonzero function u which satisfies the BC's, calculate (Lu, u) in two different ways).